



LV.0 → LV.1 · AI ENGINEER

The 3 X's that move you from **Level 0** → **1** AI engineer.

XGBoost, Explainable AI & XLA. **Plain-English** definitions, one swipe each, plus the X-named labs and the mind to follow.

EXTREME GRADIENT BOOSTING • LEARNING

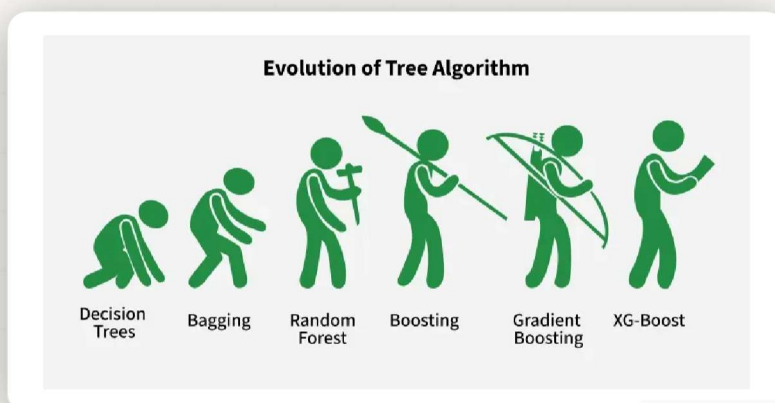
XGBoost

many weak trees • one strong model

XGBoost chains hundreds of small decision trees, each one **correcting the mistakes of the last**, into the algorithm that still wins most problems on structured, tabular data.

USED IN

- Tabular ML
- Finance & risk
- Kaggle winners



EVOLUTION OF TREE ALGORITHMS

EXPLAINABLE AI • ALIGNMENT

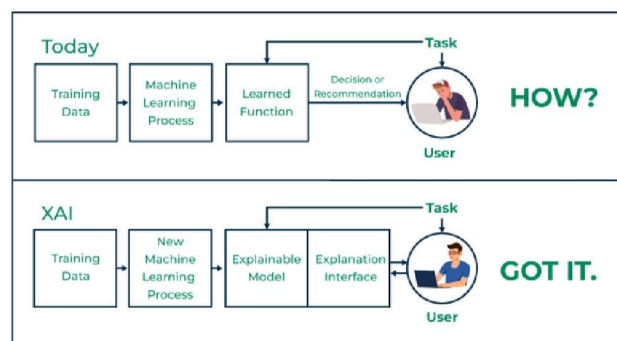
XAI

show the why • not just the what

XAI makes a model's decisions **transparent enough for a human to understand and trust**, turning black-box predictions into reasons you can audit, regulate, and act on.

USED IN

- AI governance
- Auditing & trust
- High-stakes decisions



FROM BLACK BOX TO REASONS

ACCELERATED LINEAR ALGEBRA • INFRASTRUCTURE

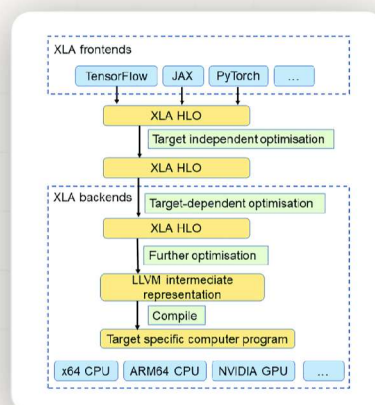
XLA

compile the math • run it faster

XLA is Google's compiler that **fuses and optimizes ML math into fast machine code** for CPUs, GPUs, and TPUs, making large-scale training and inference dramatically more efficient.

USED IN

- TensorFlow / JAX
- TPU training
- Graph optimization



FRONTENDS → OPTIMIZED BACKENDS



↓ UP NEXT • COMPANIES

The X labs.

Beyond the **3 X-words**, the X-named labs turning these ideas into real products, from frontier models to AI drug discovery.



FRONTIER AI LAB

xAI

Grok • reasoning • real-time AI



Elon Musk's frontier lab building the **Grok family of models** on its Colossus supercomputer, wired into X for real-time information.

WHAT IT IS

Frontier AI lab founded by Elon Musk.

WHY IT MATTERS

A top lab racing OpenAI & DeepMind.

WHAT THEY BUILD

Grok LLMs, reasoning & agents.

FIND THEM

x.com/xai • [@xai](https://twitter.com/xai)

OPEN-WEIGHT MODEL FAMILY

Xiaomi MiMo

open weights • reasoning • coding

Xiaomi MiMo

Xiaomi's open-source LLM family tuned for **reasoning, math, and code**, marking the phone giant's entry into frontier AI research.

WHAT IT IS

Open-weight LLM family by Xiaomi.

WHY IT MATTERS

Big-tech entry into open frontier AI.

WHAT THEY BUILD

Reasoning, math & coding models.

FIND THEM

huggingface.co/XiaomiMiMo

AI-NATIVE BIOTECH

Xaira

AI-native drug discovery

The Xaira logo, featuring the word "xaira" in a lowercase, sans-serif font, with a small square icon to the left of the text.

An AI-native biotech using **foundation models on biological data** to discover new medicines, one of the best-funded bets in AI drug discovery.

WHAT IT IS

AI-native biotechnology company.

WHY IT MATTERS

Brings frontier AI into pharma R&D.

WHAT THEY BUILD

Bio foundation models for drugs.

FIND THEM

xaira.com

AI + PHYSICS FOR SCIENCE

XtalPi

AI • quantum physics • robotics

Combines **AI, quantum physics, and lab robotics** to predict molecules and accelerate drug and materials discovery.



WHAT IT IS

AI-driven pharma & materials lab.

WHY IT MATTERS

Fuses AI, quantum & automation.

WHAT THEY BUILD

Molecular design via AI + physics.

FIND THEM

en.xtalpi.com

AGENTIC AI STARTUP

Xane AI

autonomous AI employees

xane.ai
Experience Redefined

Builds **autonomous AI employees** that execute real business tasks, sales, support, and operations, not just answer questions.

WHAT IT IS

Agentic-AI startup.

WHY IT MATTERS

Part of the rising AI-agent wave.

WHAT THEY BUILD

AI “employees” that do tasks.

FIND THEM

xane.ai • @Xane_AI



↓ UP NEXT • PEOPLE TO FOLLOW

Spotlight next.

The **high-signal, low-noise** mind worth a follow
to keep your X-stack current.



HIGH-SIGNAL, LOW-NOISE

Xavier Amatriain

LinkedIn • @xamat



A leader in **applied machine learning** whose work on recommender systems shaped modern personalization, now writing clearly about real-world AI.

WHAT THEY POST

Applied AI, GenAI & recsys explainers.

FIND THEM

LinkedIn in/xamat • amatriain.net

WHY FOLLOW

Bridges research and real products.

BEST FOR

Turning AI research into practice.

The X stack, in one card.

Save it for later. Drop a comment if there's an X-term you'd add.

01 **XGBoost**

learning

02 **XAI**

alignment

03 **XLA**

infrastructure

+ **xAI**

company

+ **Xiaomi MiMo**

company

+ **Xaira**

company

+ **XtalPi**

company

+ **Xane AI**

company

@ **Xavier Amatriain**

people



FOLLOW FOR THE FULL A TO Z

Prasanna Kulkarni

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Y Next